SP26 COMPNET — Intro to Net. and Telecom.

1 Instructor Information

Name: **Zelalem Jembre Yalew (Zee)**

Email: **zyalew@umbc.edu**

Office phone: **410-615-3535**

Office location: **ITE 329-A**

Office hours: **Mon/Wed. 1:00 - 2:00 PM**

2 Learning Resources

Schedule: **MoWe 09:30 AM - 12:30 PM & Fr 09:30 AM - 12:00 PM, Ch 114**

Weekly Schedule: **Subjected to change without notification from the professor.**

Textbook (Suggested): **Computer Networking: A Top-Down Approach 6th Ed., James Kurose**

Course discord: `https://discord.gg/Dk9ePRkhNJ`

Notebook LM: **An assistant to help with logistics**

Codes and Practice: **Github classroom repo**

Online Classroom: **https://meet.google.com/kce-jdqa-xsi**

Teaching Assistant: **Yashaswini Prashanth Kumar**

Book Appointment: `https://calendar.app.google/C6Ky17Vh2sYXUsQf6`

- I am available for consultation both in person and online. **Walk-ins** are **welcome**; however, **priority** is given **to** students with scheduled **appointments**.
  - For **in-person** meetings, please come to my office in **ITE 329-A**.
  - For **online** meetings, please join the designated **Goole** meeting room.

3 Weather Policy

If school is closed due to weather, we will hold classes online:

- **Sync**: online using the designated Google Meet or
- **Async**: recorded videos posted on blackboard.

4 Communication Policy

Discord is simpler, quicker, and preferred by me to discuss the text based questions or concerns.

When you need to send me an email, which could be sensitive/private data (e.g. doctors note), please allow 24–48 hours for a response (I am usually faster, except on holidays or when traveling).

On average, I teach over 200 students each semester. To help ensure I can respond efficiently, please follow these guidelines:

- Include your course number in the subject line (e.g., `[COMPNET]`).
- Be specific in your subject and message body.
- Provide solid info (screenshots, error messages, lab #) to reduce back-and-forth.
- Share only the relevant section where the error occurs. Avoid pasting entire long scripts.
- When in doubt, include a screenshot of the error.

Examples

Exhibit A: Fantastic Email

Subject: [COMPNET] Command to stat apache does not work

```
I am attempting Lab #2, but my docker would not run,
here is the command I am suing:

    docker exec -it web apachectl start

[screenshot of produced error]
```

Why good? Clear subject, specific problem, minimal command, screenshot, and direct question.

Exhibit B: Just Okay Email

Subject: [COMPNET] Command fail

```
I get an error when I try to start my apache.
It says "cannot reliably determine". What's wrong?

[All error output]
```

Why okay? The problem is described, but far too much error. Trim to only the relevant section.

Exhibit C: Not Productive Email

Subject: Command

```
My command keeps breaking. It works for other students.
Please help ASAP.
```

Why poor? No context (course, lab, error), and no error snippet.

5 Overview

This interactive and immersive hands-on course is tailored for professionals seeking to learn about computer networks with a focus on translating knowledge into practical applications within the workplace. Participants will gain expertise in cutting-edge tools and debugging techniques essential for addressing the dynamic demands of their roles. By bridging the gap between theory and practice, this course serves as a valuable supplement to academic qualifications and certifications.

The labs of this course witll make use of Linux, VSCode/Emacs, Github, Docker, and GL programming environment.

6 Course Objectives

The objectives of this course are to:

- Understand Internet structure, and network layer stack
- Understand Application layer protocol design and its implementation
- Analyze working of HTTP and its evolution to most recent version HTTP/3
- Analyze Transport Layer protocols and mechanisms of reliable delivery
- Understand IP addressing, subnetting and address lab and routing of packets as best effort delivery
- Design and implement simple network and study network behavior in real life.
- Become familiar with common network tools to analyze network traffic.
- Understand wireless networking

7 Course Description

This interactive hands-on course will introduce computer networks following a systems approach using experiential learning methodology. The course will start with a basic introduction of Linux and its commands and utilities. The tools and utilities will be instrumental in design, implementation and debugging of network topologies, and understanding application behavior. The course will then introduce IP addressing that plays a pivotal role in enabling communication between internetworked systems. As the Internet has become an integral part of our daily lives, the Internet structure and packet switching will be introduced next. This will be followed by understanding an overview of basic tools that empower an engineer/practitioner to understand and dissect the network operation and performance.

Studying networks basically refers to understanding the underlying protocols, often referred to as the protocol stack. The first protocol this course will cover in detail is HTTP, the web application protocol, essentially the one that runs all Internet applications. Next, we will discuss transport layer protocols in detail, namely, TCP and UDP. The end-to-end delivery has to make use of a network layer protocol that is responsible for delivering data packets across the Internet. This requires a better understanding of packet routing as well as diagnosis, and the course will cover basics of IP packet forwarding and the Internet Control Message Protocol (ICMP) protocol. Lastly, we will cover Link layer protocols which deal with communication over a connection between two machines on the network. The course will end with a basic introduction of wireless networking, namely, Wi-Fi, Bluetooth and Cellular networks, and thus completing the networking protocol stack.

Course Structure

Each week, the course includes two lecture sessions and one laboratory session. The instructor delivers the lectures and introduces key lab concepts to prepare students for the hands-on component.

During the laboratory session, depending on the topic and available time, students will begin with a brief review led by the teaching assistant and, when applicable, a pre-lab quiz. Students will then proceed to complete the assigned laboratory exercises.

The course will conclude with a term project in which students will design and implement a networked application based on the concepts and techniques learned throughout the semester.

Each topic is paired with a practical exercise or an instructor-led walkthrough designed to reinforce theoretical material. These hands-on labs provide students with a strong technical understanding of networks and allow them to gain experience with real-world network operations.

- Hands on Labs:
  - The majority of lab experiments will be done using laptops, VMs and docker instances.
  - If necessary exercises would use real hardware devices: Layer-2 switches, and Wi-Fi Access Point/Router.
  - UMBC will provide all the equipment needed to support Labs.
- Hardware requirement
  - Students can use personal laptops. The recommended specso for running labs are:
    - OS: Windows, macOS, or Linux.
    - RAM: At least 8GB of RAM (16GB recommended when not deleting other labs).
    - Processor: Minimum Intel i5 or equivalent; i7 or equivalent recommended.
    - Storage: 20GB of free space for datasets, libraries, and Docker images
    - Graphics: No dedicated GPU required, but helpful for some tasks.
  - UMBC will provide access to cloud environments or computing clusters (if available).
- Software Installation:
  - Students will be provided with a setup guide for installing Anaconda, Docker, and necessary dependencies.
  - Datasets used in the course will be downloadable, with links provided in lab labs.